

University of Science and Technology of China

Topological Rigidity under Nonnegative Scalar Curvature

Jiangcheng You

School of Mathematical Sciences, USTC

Joint work with Heng Zhang

Institute of Mathematics, AMSS, Chinese Academy of Sciences

September 10, 2026

Outline

1. Historical literature review
2. Main theorems and Motivation
3. Step 1 of the proof: topological reduction
4. Step 2 of the proof: from topology to a uniform flux gap
5. Step 3 of the proof: from the flux gap to contradiction
6. Step 4 of the proof: removing the nonparabolicity assumption

Two rigidity questions

Wang's question

Let M^3 be contractible and admit a complete metric with

$$R_g \geq 0.$$

Must

$$M \cong_{\text{diff}} \mathbb{R}^3?$$

Gromov's question

Let M_γ be the interior of a compact handlebody of genus γ and admit a complete metric with

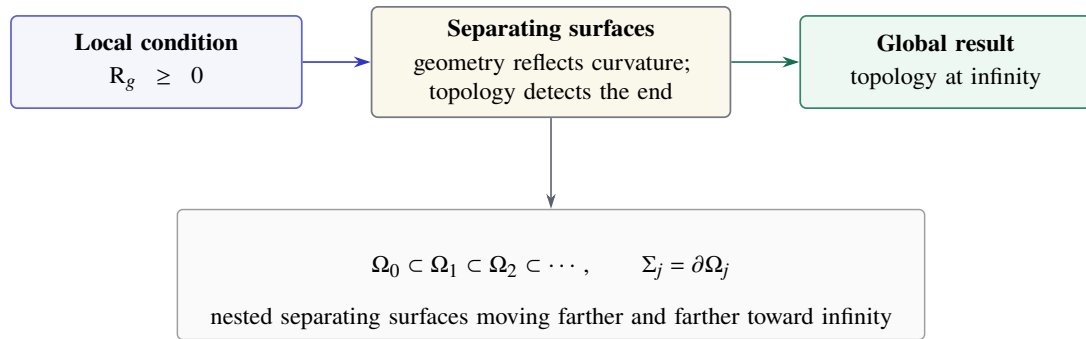
$$R_g \geq 0.$$

Must

$$\gamma \leq 1?$$

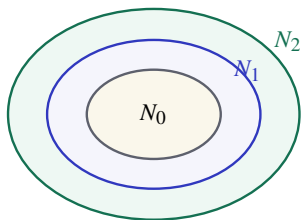
The difficulty is a mismatch of scales: **scalar curvature is local**, while the conclusion concerns **topology at infinity**.

The geometric problem



Recurring question: How can we construct separating surfaces farther and farther out whose **geometry is controlled by R_g** , while their **topology still detects the topology at infinity**?

Wang's genus-one setting [24,JDG]



$N_0 \subset N_1 \subset N_2 \subset \dots \nearrow M$
defining sequence of solid tori

Contractible genus-one manifolds

Wang considers

$$M = \bigcup_{k=0}^{\infty} N_k,$$

where each N_k is a solid torus and N_k is contractible in N_{k+1} .

But the nesting is nontrivial

Despite this homotopy triviality,

$$I(N_k, N_{k+1}) \geq 2.$$

$\pi_1(M) = 0$ does not see the topology at infinity; the **nesting of the solid tori** does.

Wang: topology at infinity versus the stable limit [24,JEMS]

Assume that M is contractible, $R_g > 0$, $\pi_1^\infty(M) = 0$, and $M \not\cong \mathbb{R}^3$.

Topology produces persistent meridians

Choose an exhaustion by handlebodies N_j and a family R_k with Property (H).

meridians of $N_j \implies$ stable minimal disks

$$\implies \mathcal{L}_j \longrightarrow \mathcal{L} = \bigcup_{t \in \Lambda} L_t.$$

For every fixed R_k , and all sufficiently large j ,

$$\mathcal{L}_j \cap \partial R_k$$

contains a **meridian** of R_k .

The vanishing property of $\mathcal{L}$

Wang's Vanishing Property gives, for k large,

$$\gamma \subset L_t \cap \partial R_k \implies [\gamma] = 0 \in \pi_1(\partial R_k).$$

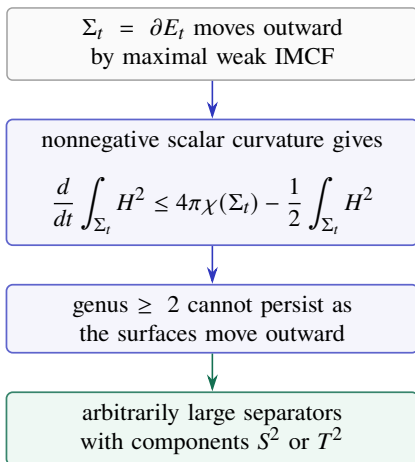
Failure of this property would produce a nontrivial element of $\pi_1^\infty(M)$.

Hence

$$\pi_1^\infty = 0 + \text{stability} + R_g > 0 \implies \text{Vanishing Property.}$$

Topology forces meridians in $\mathcal{L}_j$, while the stable limit forces them to vanish: **contradiction**.

Chodosh–Lai–Xu: the separator itself move outward [26,CPAM]



Green levels: analytic separating surfaces (CM [25,arXiv],YZ [26,arXiv])

Colding–Minicozzi: analytic control

For a positive Green function G , set

$$b = G^{-1}, \quad \Sigma_r = \{b = r\}.$$

Bochner–Gauss identities and Gauss–Bonnet relate the geometry and topology of Σ_r to differential inequalities for level-set energies.

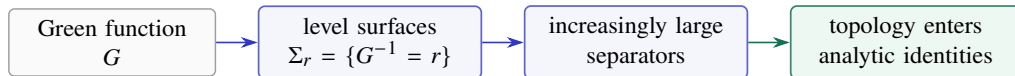
Yan–Zhu: increasingly large separators

If

$$G(x) \rightarrow 0 \quad (x \rightarrow \infty),$$

then $b = G^{-1}$ is proper.

Hence the level sets Σ_r move outward and exhaust the end as $r \rightarrow \infty$.



T. Colding–W. Minicozzi II, arXiv:2501.17947; Z. Yan–X. Zhu, arXiv:2604.08285.

The common feature

One outward geometric family is asked to do several jobs

interact with R_g

remain compact
/ controlled

reach arbitrary
scales

detect topology
at infinity

Weak IMCF

Bounded geometry supplies the large-scale control needed to analyze proper, sweeping, and escaping behavior.

Global Green levels

The same Green levels must also exhaust the manifold. This requires decay/properness of G and large-scale gradient control, typically supplied by a Ricci lower bound.

Can Green-function levels keep their strong scalar-curvature calculus without also being responsible for exhausting the entire manifold?

Roadmap

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Main theorems

Contractible case

If (M^3, g) is complete, contractible, and $R_g \geq 0$, then

$$M \cong_{\text{diff}} \mathbb{R}^3.$$

Open-handlebody case

If the interior M_γ of a compact handlebody admits a complete metric with $R_g \geq 0$, then

$$\gamma \leq 1.$$

Yan–Zhu: the argument in nonparabolic setting [26,arXiv]

Topology + CM calculus

If high genus persists along the Green levels, Gauss–Bonnet creates a definite gap in the Colding–Minicozzi inequalities:

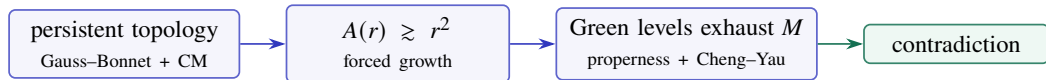
$$\text{persistent high genus} \implies A(r) \gtrsim r^2.$$

Thus topology forces increasingly strong **growth along the Green levels**.

Global control at infinity

In the nonparabolic setting, take a global minimal positive Green function G .

- Decay at infinity $G(x) \rightarrow 0$ gives $b = G^{-1}$ is proper, so $\Sigma_r = \{b = r\}$ themselves move to infinity.
- A Ricci lower bound gives Cheng–Yau gradient control, which is incompatible with the forced quadratic growth.



Yan–Zhu make **one global Green-level family do two jobs**: carry the CM analysis and simultaneously exhaust the manifold.

Motivation: keep the local computations, change the exhaustion

Colding–Minicozzi

Yan–Zhu

our paper

Common local engine: $G > 0 \rightarrow b = G^{-1} \rightarrow \Sigma_r = \{b = r\}$
Bochner–Gauss + Gauss–Bonnet $\Rightarrow$ topology forces Green-level growth

What we keep

The same local Green-level mechanism:

high genus $\Rightarrow$ Gauss–Bonnet gap $\Rightarrow A(r) \gtrsim r^2$.

This is the analytic engine of the argument.

What we change

We no longer require the Green levels themselves to exhaust the manifold.

Instead, we separate

r : local Green-level motion,
 τ : global exhaustion.

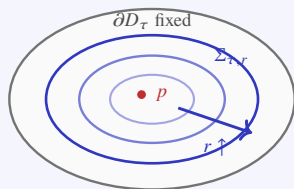
Decouple the two jobs: keep the Colding–Minicozzi analysis local, and introduce an independent outer exhaustion.

Turning point: decouple the two motions

For each exhaustion domain D_τ , let

$$G_\tau = G_{D_\tau}(p, \cdot), \quad b_\tau = G_\tau^{-1}, \quad \Sigma_{\tau,r} = \{b_\tau = r\}.$$

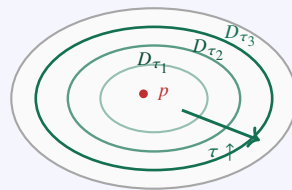
1. Fix τ : move the Green level



$$r \rightarrow \infty \implies \Sigma_{\tau,r} \rightarrow \partial D_\tau.$$

Here D_τ is fixed. This is the parameter used in the **Colding–Minicozzi level-set calculation**.

2. Increase τ : move the domain



$$D_{\tau_1} \subset D_{\tau_2} \subset \cdots, \quad D_\tau \nearrow M.$$

Here the **Dirichlet domain itself moves**. This is the parameter that carries the argument **globally toward infinity**.

Different choices of green function

Nonparabolicity does not give the properness we need

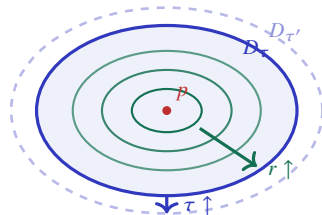
Nonparabolicity gives a minimal positive Green function G_M , but not necessarily $G_M(x) \rightarrow 0$ when $x \rightarrow \infty$. Hence G_M^{-1} need not be proper, so its levels may fail to detect the topology at infinity.

Localize

Choose an exhaustion $D_\tau \uparrow M$ and let G_τ be the Dirichlet Green function on D_τ . Then

$$G_\tau = 0 \text{ on } \partial D_\tau \implies b_\tau := G_\tau^{-1} \rightarrow \infty.$$

Thus the Green levels $\Sigma_{\tau,r} := \{b_\tau = r\}$ sweep from p toward ∂D_τ as $r \uparrow \infty$.



r : move Green levels

τ : enlarge the outer domain

Local Green functions provide the moving surfaces; the global Green function is used only for comparison.

Inner motion: From the CM growth to boundary flux

What the inner analysis gives

For every sufficiently large Green level in D_τ ,

$$A_\tau(r) \geq c_* r^2,$$

with $c_* > 0$ independent of τ .

Boundary limit

Set $q_\tau = -\partial_{\nu_\tau} G_\tau$. Then

$$\frac{A_\tau(r)}{r^2} \longrightarrow \int_{\partial D_\tau} q_\tau^2 dA \quad (r \rightarrow \infty).$$

Thus the quadratic CM growth becomes a **uniform positive boundary flux**.

$$\frac{A_\tau(r)}{r^2} \geq c_*$$

$r \rightarrow \infty$
levels approach ∂D_τ

$$\int_{\partial D_\tau} q_\tau^2 dA \geq c_*$$

Outer motion: boundary flux forces Robin growth

Now let the outer domain move

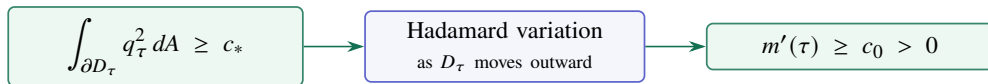
For each D_τ , write

$$m(\tau) := m_{D_\tau}(p)$$

for the **Robin constant** of the Dirichlet Green function G_τ .

Hadamard variation expresses its change in terms of the boundary flux:

$$m'(\tau) = \frac{1}{4\pi} \int_{\partial D_\tau} (\text{boundary speed}) q_\tau^2 dA.$$



The flux obtained from the inner r -analysis now becomes **uniform growth in the outer parameter τ** .

The contradiction

What the outer motion gives

From the uniform boundary flux,

$$m'(\tau) \geq c_0 > 0$$

on regular intervals.

Critical outer levels can only add **nonnegative jumps**.

Hence

$$m(\tau) \longrightarrow +\infty \quad (\tau \rightarrow \infty).$$

What nonparabolicity gives

If M is nonparabolic, there exists a **global minimal positive Green function**. Let $m_M(p)$ be its Robin constant.

Domain monotonicity gives

$$m(\tau) = m_{D_\tau}(p) \leq m_M(p) < \infty.$$

Therefore the previous growth is impossible.



The global Green function provides a finite Robin barrier. Its positive levels need not exhaust M .

From nonparabolic metric to general metric

Fixed-metric analytic theorem

For a complete, one-ended, **nonparabolic** 3-manifold,

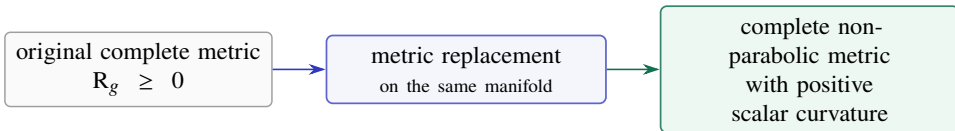
$$R_g \geq 0, \quad b_1(M) < \infty,$$

the Green-function argument forces low-genus separators at infinity.

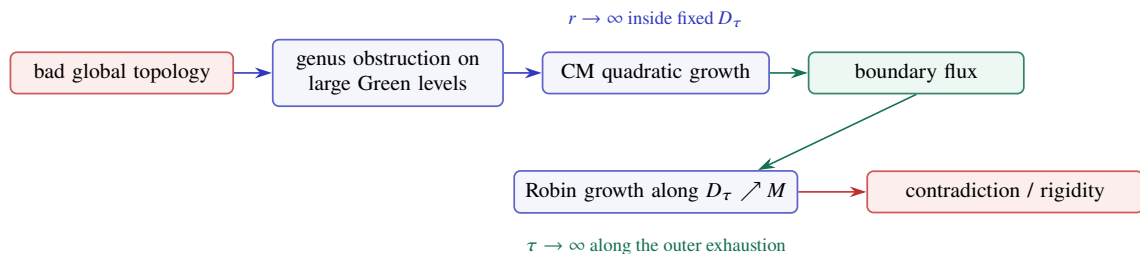
But our rigidity theorem assumes only

$$\text{completeness} \quad + \quad R_g \geq 0.$$

So one more step is needed: **remove the nonparabolicity assumption without changing the topology.**



Proof roadmap: two-stage transfer from topology to infinity



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The low-genus separator property

Admissible separator

A smooth precompact domain $U \Subset M$ such that

- U is connected;
- $M \setminus U$ is connected;
- ∂U is smooth, compact and connected.

Property (LG)

Every compact $K \Subset M$ is contained in an admissible separator U satisfying

$$\text{genus}(\partial U) \leq 1.$$

Property (LG) is the exact interface between the analytic argument and the topological rigidity theorem.

Why (LG) settles both rigidity questions

Contractible case

Repeatedly apply (LG) to obtain a nested exhaustion by connected surfaces of genus 0 or 1.

- sphere exhaustion $\Rightarrow M \cong \mathbb{R}^3$ (Husch–Price);
- torus exhaustion $\Rightarrow M \cong \mathbb{R}^3$ (Wang; cf. Yan–Zhu, Lemma 2.4).

Open handlebody M_γ

A fixed compact core has end

$$\Sigma_\gamma \times (0, \infty).$$

Any admissible ∂U enclosing it maps to Σ_γ with degree ± 1 , so

$$\gamma \leq \text{genus}(\partial U) \leq 1.$$

From now on, geometry and analysis have one job: prove (LG).

Negating (LG): what this means for the Green levels

A fixed topological obstruction

Assume (LG) fails. Then there exists a compact set $K_0 \Subset M$ such that

$$K_0 \subset U, \quad U \text{ admissible} \implies \text{genus}(\partial U) \geq 2.$$

Apply the obstruction to Green levels

For each Dirichlet Green function G_τ , recall

$$b_\tau := G_\tau^{-1}, \quad U_{\tau,r} := \{b_\tau < r\}, \quad \Sigma_{\tau,r} := \{b_\tau = r\}.$$

For every regular value r , $\partial U_{\tau,r} = \Sigma_{\tau,r}$. Hence, once we prove $K_0 \subset U_{\tau,r}$ and $U_{\tau,r}$ is admissible, the contradiction hypothesis gives

$$\boxed{\text{genus}(\Sigma_{\tau,r}) \geq 2.}$$

The topological task is concrete: **make the Green superlevel domains contain K_0 and prove that their regular boundaries are admissible separators.**

Build a controlled outer exhaustion

To test the fixed obstruction K_0 with Green levels, we first need a nested family of bounded domains on which the Dirichlet Green functions can be compared.

Filled Morse exhaustion

Choose a proper Morse function

$$\rho : M \rightarrow [0, \infty), \quad |\nabla \rho| \leq L.$$

If C_τ is the component of $\{\rho < \tau\}$ containing p , fill all precompact complementary components:

$$D_\tau := C_\tau \cup \{\text{precompact complementary components}\}.$$

For regular τ , there holds $D_\tau \Subset M$, both D_τ and $M \setminus D_\tau$ are connected, and

$$D_\sigma \subset D_\tau \quad (\sigma < \tau), \quad D_\tau \uparrow M.$$

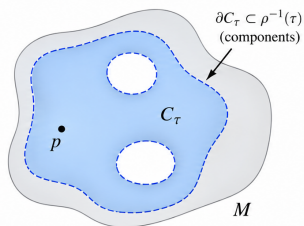
Moreover, the unit outward normal vector and the outward normal velocity satisfy

$$v_\tau = \frac{\nabla \rho}{|\nabla \rho|}, \quad V_\tau = \frac{1}{|\nabla \rho|} \geq L^{-1}.$$

The parameter τ gives a nested outer family with uniform control of its outward motion.

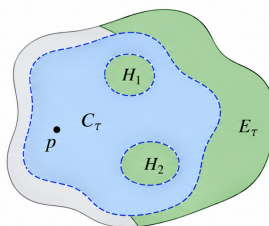
Construction of the filled exhaustion D_τ

(A) The sublevel set $\{\rho < \tau\}$
and its component C_τ



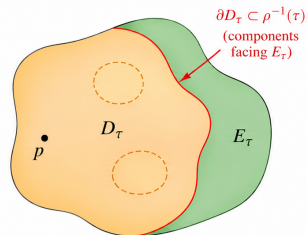
C_τ is the component of $\{\rho < \tau\}$ containing p .

(B) The complementary components
of $M \setminus C_\tau$



$$M \setminus C_\tau = E_\tau \sqcup H_1 \sqcup H_2$$

(C) The filled domain $D_\tau = M \setminus E_\tau$



$$D_\tau = C_\tau \cup H_1 \cup H_2 = M \setminus E_\tau$$

One threshold makes the Green levels admissible

Enlarge the fixed obstruction once and for all:

$$K \ni K_0 \cup S \cup \{p\},$$

where S is used to determine whether the boundary of a connected domain is connected. Since $D_\tau \uparrow M$, choose a regular τ_0 such that

$$K \Subset D_{\tau_0}.$$

A threshold independent of the outer domain

Since $G_{\tau_0} > 0$ on $K \setminus \{p\}$, choose $r_* > 0$ so that

$$G_{\tau_0} > r_*^{-1} \quad \text{on } K \setminus \{p\}.$$

Domain monotonicity of green functions gives

$$K \subset U_{\tau,r} \quad (\tau \geq \tau_0, r \geq r_*).$$

For regular $\tau \geq \tau_0$ and regular $r \geq r_*$, both $U_{\tau,r}$ and $M \setminus U_{\tau,r}$ are connected, and $S \subset U_{\tau,r}$ implies $\Sigma_{\tau,r}$ is connected. Hence $U_{\tau,r}$ is an admissible separator containing K_0 . By the failure of (LG),

$$\text{genus}(\Sigma_{\tau,r}) \geq 2.$$

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From the genus gap to a flux gap

The previous step produced constants τ_0 and r_* such that, for every sufficiently large regular τ and every regular $r \geq r_*$,

$$\Sigma_{\tau,r} \text{ is connected and } \text{genus}(\Sigma_{\tau,r}) \geq 2.$$

Fix such a τ . Write

$$G_\tau = G_{D_\tau}(p, \cdot), \quad b_\tau = G_\tau^{-1}, \quad q_\tau = -\partial_{\nu_\tau} G_\tau.$$

Our goal in this section is

$$\int_{\partial D_\tau} q_\tau^2 dA \geq c_* > 0,$$

where c_* is independent of the outer parameter τ .

$$\text{genus}(\Sigma_{\tau,r}) \geq 2 \implies \text{uniform boundary flux gap.}$$

Why quadratic growth is the right analytic target

Fix one sufficiently large regular τ and define

$$A_\tau(r) := r^{-2} \int_{\Sigma_{\tau,r}} |\nabla b_\tau|^2 dA.$$

Then

$$\frac{A_\tau(r)}{r^2} = \int_{\{G_\tau=r^{-1}\}} |\nabla G_\tau|^2 dA.$$

As $r \rightarrow \infty$, the Green level approaches ∂D_τ , hence

$$\lim_{r \rightarrow \infty} \frac{A_\tau(r)}{r^2} = \int_{\partial D_\tau} q_\tau^2 dA.$$

Therefore it is enough to prove $A_\tau(r) \geq c_* r^2$ with $c_* > 0$ independent of τ .

The CM identity: topology becomes positive forcing

Suppress τ for one fixed outer domain. The exact Colding–Minicozzi identity can be written as

$$rA'(r) = A(r) + \frac{1}{2r} \int_0^r (S_1(s) + B_2(s) - 2\kappa(s)) ds.$$

Here

$$S_1(r) = \int_{\Sigma_r} R_g dA \geq 0, \quad B_2(r) = \int_{\Sigma_r} \frac{|B|^2}{4r^2 |\nabla b|^2} dA_g \geq 0, \quad \kappa(r) = \int_{\Sigma_r} K_{\Sigma_r} dA.$$

For every regular $r \geq r_*$,

$$\text{genus}(\Sigma_r) \geq 2 \implies \kappa(r) = 2\pi\chi(\Sigma_r) \leq -4\pi,$$

so

$$-2\kappa(r) \geq 8\pi.$$

$$\text{genus gap} \xrightarrow{\text{Gauss–Bonnet}} \text{persistent positive forcing for } A.$$

Step 2.1: The genus gap enters the CM computational framework

Put $\beta := b_1(M)$. For every regular level, the component estimate and Gauss–Bonnet give

$$\kappa(r) = 4\pi \sum_{j=1}^{N(r)} (1 - g_j) \leq 4\pi(\beta + 1).$$

Combining this estimate with the estimate $\kappa(r) \leq -4\pi$ arising from the topological obstruction, we obtain that for every regular $r \geq r_*$,

$$\int_0^r (-2\kappa(s)) ds \geq -8\pi(\beta + 1)r_* + 8\pi(r - r_*) = 8\pi r - 8\pi(\beta + 2)r_*.$$

Hence

$$rA'(r) \geq A(r) + 4\pi - \frac{4\pi(\beta + 2)r_*}{r}.$$

Define $R_0 := 2(\beta + 2)r_*$, then, for $r \geq R_0$,

$$rA'(r) \geq A(r) + 2\pi.$$

With $r_1 := 2R_0$,

$$\left(\frac{A}{r}\right)' \geq \frac{2\pi}{r^2} \quad \implies \quad A(r_1) \geq 2\pi, \quad a(r_1) := \frac{r_1 A'(r_1)}{A(r_1)} \geq 1.$$

Step 2.2: derive a integral Riccati inequality

Define the logarithmic growth exponent

$$a(r) := \frac{rA'(r)}{A(r)}.$$

Recall $B = \text{Hess}(b^2) - 2|\nabla b|^2 g$, then B is trace-free and

$$\int_{\Sigma_r} B(\nu, \nu) dA = 2r^2 a(r) A(r).$$

Using Cauchy–Schwarz on Σ_r ,

$$\begin{aligned} B_2(r) &= \int_{\Sigma_r} \frac{|B|^2}{4r^2 |\nabla b|^2} dA_g \geq \frac{3}{8r^2} \int_{\Sigma_r} \frac{B(\nu, \nu)^2}{|\nabla b|^2} dA_g \\ &\geq \frac{3}{8r^2} \frac{\left(\int_{\Sigma_r} B(\nu, \nu) dA_g \right)^2}{\int_{\Sigma_r} |\nabla b|^2 dA_g} = \frac{3}{2} a(r)^2 A(r). \end{aligned}$$

The logarithmic growth rate admits such the control: $\frac{B_2(r)}{A(r)} \geq \frac{3}{2} a(r)^2$.

Step 2.2: derive an integral Riccati inequality

Set

$$W := |\nabla b^2|, \quad V := \frac{b^{-2}}{A(b)} \nabla W.$$

The weighted Bochner–Gauss computation gives, for a.e. regular $r \geq r_1$,

$$r \int_{\Sigma_r} \frac{\operatorname{div} V}{|\nabla b|} dA = \frac{S_1(r) + B_2(r) - 2\kappa(r)}{A(r)} + 2(1 - a(r)^2).$$

Using the geometric conditions together with the growth estimate for $a(r)$, we obtain

$$r \int_{\Sigma_r} \frac{\operatorname{div} V}{|\nabla b|} dA \geq 2 - \frac{1}{2}a(r)^2.$$

Moreover, $\partial_\nu W = \operatorname{Hess}(b^2)(\nu, \nu)$ implies

$$\int_{\Sigma_r} \langle V, \nu \rangle dA = 2a(r) + 2.$$

Thus, for regular $r_1 \leq s_1 < s_2$, Gauss–Green on $\{s_1 < b < s_2\}$ gives

$$a(s_2) - a(s_1) \geq \int_{s_1}^{s_2} \left(1 - \frac{a(r)^2}{4}\right) \frac{dr}{r}.$$

By continuity of a , the inequality extends to arbitrary $r_1 \leq s_1 < s_2$.

Step 2.3: Riccati comparison gives the quadratic growth

The comparison equation is

$$r\varphi'(r) = 1 - \frac{\varphi(r)^2}{4}, \quad \varphi(r_1) = 1.$$

Since $a(r_1) \geq 1$, the integral comparison principle gives

$$a(r) \geq \varphi(r) := 2 \frac{3r - r_1}{3r + r_1} \quad (r \geq r_1).$$

Now integrate the logarithmic growth:

$$\log \frac{A(r)}{A(r_1)} = \int_{r_1}^r a(s) \frac{ds}{s} \geq 2 \log \frac{r}{r_1} - 4 \log \frac{4r}{3r + r_1}.$$

Thus we obtain the growth estimate for $A(r)$. Combining this with $A(r_1) \geq 2\pi$ and $(3r + r_1)/(4r) \geq 3/4$,

$$A(r) \geq c_* r^2, \quad c_* := \frac{2\pi}{r_1^2} \left(\frac{3}{4}\right)^4 = \frac{81\pi}{128r_1^2}.$$

The constants r_1 and c_* depend only on $b_1(M)$ and r_* , hence are independent of τ .

Step 2.4: returns to the boundary

Restore the outer-domain subscript. For every sufficiently large regular τ ,

$$A_\tau(r) \geq c_* r^2 \quad (r \geq r_1),$$

with the same constant $c_* > 0$.

On $\Sigma_{\tau,r} = \{b_\tau = r\} = \{G_\tau = r^{-1}\}$, we have

$$\frac{A_\tau(r)}{r^2} = \int_{\{G_\tau = r^{-1}\}} |\nabla G_\tau|^2 dA.$$

Boundary regularity gives

$$\int_{\partial D_\tau} q_\tau^2 dA = \lim_{r \rightarrow \infty} \frac{A_\tau(r)}{r^2} \geq c_*.$$

genus gap $\implies$ quadratic growth $\implies$ uniform boundary flux gap.

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The outer parameter now closes the proof

Return to Hadamard's formula:

$$\frac{d}{d\tau} m_{D_\tau}(p) \geq \frac{1}{4\pi L} \int_{\partial D_\tau} q_\tau^2 dA \geq \frac{c_*}{4\pi L}$$

on every regular interval. Critical Morse values contribute only nonnegative jumps.

Therefore

$$m_{D_\tau}(p) \longrightarrow +\infty \quad (\tau \rightarrow \infty).$$

But the global minimal Green function gives

$$m_{D_\tau}(p) \leq m_M(p) < \infty.$$

Contradiction.

Hence (LG) holds. The main Green-function argument is finished.

The fixed-metric theorem

Nonparabolic low-genus separators

Let (M^3, g) be complete, orientable, one-ended, and nonparabolic. If

$$R_g \geq 0, \quad b_1(M) < \infty,$$

then M satisfies (LG).

Where topology enters

One-endedness and finite b_1 force sufficiently large Green levels to be admissible; failure of (LG) then forces their genus to be at least two.

Where nonparabolicity enters

Only at the final line:

$$m_{D_\tau}(p) \leq m_M(p) < \infty.$$

No behavior of G_M at infinity is used.

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2. Main theorems and Motivation
3. Step 1 of the proof: topological reduction
4. Step 2 of the proof: from topology to a uniform flux gap
5. Step 3 of the proof: from the flux gap to contradiction
- 6. Step 4 of the proof: removing the nonparabolicity assumption**

The remaining task

The fixed-metric theorem applies to a complete, one-ended, *nonparabolic* metric with nonnegative scalar curvature. But the original rigidity statements assume only

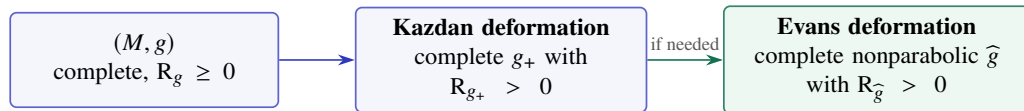
$$R_g \geq 0,$$

with no nonparabolicity hypothesis.

Can we replace the metric, without changing the manifold, so that the fixed-metric theorem becomes applicable?

The nonflat case: two deformations with separate jobs

Assume $\text{Ric}_g \not\equiv 0$. We use two metric deformations with different purposes:



Kazdan's job

Create a *strict* scalar-curvature:

$$R_{g_+} > 0.$$

Evans' job

If necessary, deform the metric at infinity so that it becomes *nonparabolic*.

Kazdan deformation: make the scalar curvature positive

Assume $R_g \geq 0$, $\text{Ric}_g \not\equiv 0$. Recall the Rayleigh quotient

$$\mu_1(\Omega, h) = \inf_{0 \neq v \in H^1(\Omega)} \frac{\int_{\Omega} (c_n |\nabla v|_h^2 + R_h v^2) dV_h}{\int_{\Omega} v^2 dV_h}. \quad (1)$$

Step 1. Produce a positive eigenvalue

For $\mathcal{L}_h := -8\Delta_h + R_h$, find $g_0 \sim g$ and $\Omega \Subset M$ such that

$$\mu_1(\Omega, g_0) > 0.$$

If $R_g \not\equiv 0$, take $g_0 = g$; if $R_g \equiv 0$, apply a compactly supported perturbation to the metric: $g_t = g - t\eta \text{Ric}_g$, then

$$\mu'_1(0) = \frac{1}{\text{Vol}_g(\Omega)} \int_{\Omega} \eta |\text{Ric}_g|^2 dV_g > 0.$$

Step 2. Kazdan's criterion's output

Kazdan's criterion gives a bounded positive function w with $0 < c_0 \leq w \leq C_0$ and $\mathcal{L}_{g_0} w > 0$ on M .

Kazdan deformation: make the scalar curvature positive.

Step 3. Conformal deformation

Set

$$g_+ = w^4 g_0.$$

Then

$$R_{g_+} = w^{-5} \mathcal{L}_{g_0} w > 0.$$

Since $g_0 \sim g$ and w is bounded above and below,

$$g_+ \sim g \implies g_+ \text{ is complete.}$$

Why do we need this deformation?

The next Evans deformation may lower scalar curvature on a compact transition region K .

With only $R_g \geq 0$, there is no room for such a loss; but now

$$R_{g_+} > 0 \implies \min_K R_{g_+} > 0.$$

This positive margin can absorb the error introduced by the Evans deformation.

Evans deformation

Now suppose (M, g_+) is still parabolic. One-endedness gives a proper exterior Evans potential

$$\Delta_{g_+} u = 0, \quad u|_{\partial E} = 0, \quad u(x) \longrightarrow \infty \quad \text{at infinity.}$$

Choose regular values $0 < a < b$ and a smooth nondecreasing function

$$\eta : [0, \infty) \rightarrow [0, 1], \quad \eta = 0 \text{ on } [0, a], \quad \eta = 1 \text{ on } [b, \infty).$$

Set

$$F_\varepsilon(t) = 1 + \varepsilon \int_0^t \eta(s) ds,$$

Define the global conformal factor

$$\phi = \begin{cases} F_\varepsilon(u), & \text{on } E, \\ 1, & \text{on } M \setminus E, \end{cases} \quad \boxed{\widehat{g} = \phi^4 g_+}.$$

Why capacity? Positive capacity of a compact set is equivalent to nonparabolicity (Grigor'yan). The linear growth of F_ε at infinity is chosen precisely to produce such positive capacity.

Evans deformation: preserve positivity and completeness

Scalar curvature

$$R_{\widehat{g}} = \phi^{-5}(-8\Delta_{g_+}\phi + R_{g_+}\phi).$$

Since

$$\Delta_{g_+}\phi = F''_{\varepsilon}(u)|\nabla u|^2,$$

the extra term is supported only on the compact band $\{a \leq u \leq b\}$. For ε small, it is dominated by $R_{g_+} > 0$.

Completeness

Since

$$\phi \geq 1,$$

we have

$$\widehat{g} = \phi^4 g_+ \geq g_+.$$

Thus divergent curves cannot become shorter, so $\widehat{g}$ remains complete.

For sufficiently small ε , $\widehat{g}$ is complete, $R_{\widehat{g}} > 0$.

Evans deformation: create nonparabolicity

Target

To prove $(M, \widehat{g})$ is nonparabolic, it is enough to show that some compact set has positive capacity. For a regular value $T > b$, define

$$I_T := \int_b^T F_\varepsilon(s)^{-2} ds, \quad h_T(x) := \frac{1}{I_T} \int_{u(x)}^T F_\varepsilon(s)^{-2} ds;$$
$$K_b := (M \setminus E) \cup \{u \leq b\}, \quad \Omega_T := (M \setminus E) \cup \{u < T\},$$

It suffices to prove

$$\lim_{T \rightarrow \infty} \text{Cap}_{\widehat{g}}(K_b; \Omega_T) = \lim_{T \rightarrow \infty} \int_{\{b < u < T\}} |\nabla h_T|_{\widehat{g}}^2 dV_{\widehat{g}} > 0.$$

Why choose h_T this way?

We seek h_T in the form $h = H(u)$ as a suitable representative for computing the capacity. The equation $\Delta_{\widehat{g}} h = 0$ reduces to $(F_\varepsilon^2 H')' = 0$, hence

$$H'(t) \propto F_\varepsilon(t)^{-2}.$$

Imposing $H(b) = 1, H(T) = 0$, gives the form used above.

Evans deformation: create nonparabolicity

Then the coarea formula gives

$$\text{Cap}_{\widehat{g}}(K_b; \Omega_T) = \int_{\{b < u < T\}} |\nabla h_T|_{\widehat{g}}^2 dV_{\widehat{g}} = \text{const} \int_b^T F_{\varepsilon}(t)^2 H'_T(t)^2 dt = \boxed{\frac{\text{const}}{I_T}}.$$

Finish the verification

For $t \geq b$, $F_{\varepsilon}(t) = F_{\varepsilon}(b) + \varepsilon(t - b)$, so

$$I_T \longrightarrow \int_b^{\infty} F_{\varepsilon}(t)^{-2} dt = \frac{1}{\varepsilon F_{\varepsilon}(b)} < \infty.$$

Therefore

$$\boxed{\text{Cap}_{\widehat{g}}(K_b) = \frac{J}{I_{\infty}} = J\varepsilon F_{\varepsilon}(b) > 0} \quad \implies \quad \boxed{(M, \widehat{g}) \text{ is nonparabolic.}}$$

Now the fixed-metric theorem applies

For the nonflat case we have constructed, on the *same manifold* M , a metric $\widehat{g}$ satisfying

$$\widehat{g} \text{ complete}, \quad R_{\widehat{g}} > 0, \quad (M, \widehat{g}) \text{ nonparabolic.}$$

The hypotheses

$$\text{orientable}, \quad \text{one-ended}, \quad b_1(M) < \infty$$

are topological and therefore unchanged.

The fixed-metric theorem applied to $(M, \widehat{g})$ gives

M satisfies (LG).

The conclusion concerns the manifold, not the chosen metric.

The flat case is separate and elementary

If $\text{Ric}_g \equiv 0$, then in dimension three $\text{Rm}_g \equiv 0$, so g is flat.

Contractible manifold

Complete + simply connected + flat gives

$$(M, g) \cong (\mathbb{R}^3, g_{\text{eucl}}).$$

Hence

$$M \cong_{\text{diff}} \mathbb{R}^3.$$

Open handlebody

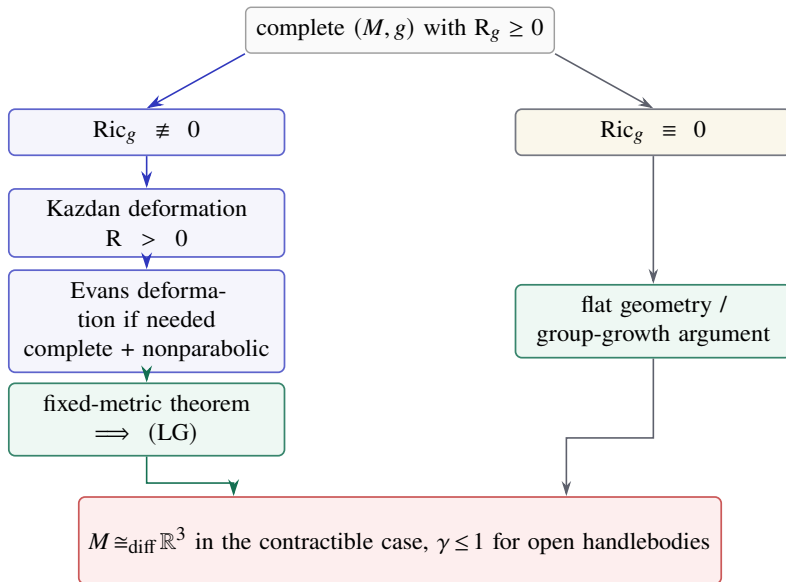
For M_γ ,

$$\pi_1(M_\gamma) \cong F_\gamma.$$

By the Euclidean packing lemma, the fundamental group of a complete flat 3-manifold, if finitely generated, has polynomial growth (of degree at most 3), whereas F_γ has exponential growth for $\gamma \geq 2$. Therefore

$$\gamma \leq 1.$$

Putting Step 4 together



Thank you

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